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Nutrition REVIEW

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Current Evidence on Common Dietary Supplements for Sleep Quality

Abstract: *Poor sleep quality impacts nearly 70 million adults in the United States, resulting in nearly 1 in 5 adults regularly utilizing some sleep aid, either prescription or over-the-counter. Common dietary supplements utilized to improve sleep quality include nitrates, melatonin, magnesium, zinc, vitamin D, and L-theanine. While underlying physiologic mechanisms support the potential impact these compounds have on sleep quality, evidence from clinical trials varies widely. Melatonin and magnesium are two of the more widely researched supplements, which have numerous studies, with conflicting results. Other compounds such as nitrates, zinc, vitamin D, and L-theanine are less supported. Additional research is needed in order to more appropriately recommend these dietary supplements for sleep improvements.*

Keywords: sleep quality; dietary supplements

A quick internet search on "sleep supplements" will yield thousands of results ranging from articles listing the top ten natural remedies to thousands of supplements for sale. It is estimated that up to 70 million people in the United States suffer from some type of sleep disorder, ranging from insomnia to narcolepsy.[1] Sleep disorders have negative impacts on mental and physical health, where it has been observed psychiatric disorders are three times more prevalent among those with sleep complaints, compared to those without complaints.[2] In addition, sleep disruptions can increase production of stress hormones like adrenaline and cortisol and increase heart rate and blood pressure.[1] Unsurprisingly, approximately 18% of adults regularly use sleep medication, both prescription and over-the-counter.[3] While many dietary supplements are advertised to promote sleep, evidence in support of their efficacy is limited. This article describes recent clinical trial evidence on nitrates, melatonin, magnesium, zinc, vitamin D, and L-theanine.

> "The supplement group exhibited enhanced sleep quality, reduced sleep latency, and decreased sleep disturbances."

Nitrates

Nitrates are molecules found in a number of foods and are also naturally occurring in the body. Nitrate medications aid in widening blood vessels; however, nitrates in various foods have different impacts on the body. In processed meats, these natural nitrates, converted to nitrites, can be potentially carcinogenic, while nitrates in vegetables, which also contain antioxidants, have been

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associated with lower rates of cancer. With regards to sleep, nitrates potentially impact sleep through increased nitric oxide, a molecule that regulates blood flow, oxygen delivery, and vascular function, to improve sleep. This may also aid in blood pressure regulation, which could also be beneficial for sleep.

Very limited evidence is available with regards to the impact of nitrate supplementation and sleep quality. In one study, healthy men were randomized to receive either a two ounce fruit and vegetable nitrate derived supplement or prune juice. The nitrate supplement group had significantly improved plasma nitrates and nitrites and decreased diastolic pressure; however, sleep quality was not impacted in either group.[4] Another study provided beetroot juice, which is rich in nitrates, to male athletes, in attempt to improve sleep to address athletic performance. The male athletes (n = 30, mean age 20.7 years) were randomized to one of three groups, 100 milliliters of beetroot juice, a placebo, or control.[5] There was a statistically significant improvement in sleep quality among the beetroot juice group, but not in the other groups and sleep quality scores in the placebo and control groups were similar.[5] The contradictory evidence available on nitrate supplementation yields limited clinical impact. Further research is needed for more conclusive evidence.

Melatonin

Melatonin is a hormone produced by the pineal gland in the brain and plays a critical role in regulation of the sleep-wake cycle. The pineal gland begins production of melatonin in response to darkness, preparing the body for sleep. As levels gradually increase, melatonin promotes relaxation and reduced alertness. During sleep, melatonin levels remain elevated, reducing the frequency of night awakenings. In addition, melatonin decreases body temperature, which aids in the onset of sleep. Melatonin supplements are synthetically produced in laboratory settings. These supplements are often utilized to manage sleep-related issues and have been purported to regulate sleep patterns, reduce time to fall asleep, improve sleep quality, and alleviate sleep disorders such as insomnia. Melatonin supplementation has been promoted for shift workers in particular, who have irregular sleep-wake schedules due to working at night.

A larger body of evidence exists on the impacts of melatonin on sleep quality. Of the seven identified randomized control trials,[6-12] five demonstrated statistically significant improvements in sleep quality, in comparison to a placebo-control.[6,8,9,11,12] Within this evidence is a degree of heterogeneity, primarily in melatonin doses, where 3 studies provided 3 mg doses[6,7,10] and 4 studies provided 10 mg doses trials.[8,9,11,12] In addition, length of intervention also varied from 30 days[6] to 6 months.[10] This wide range in doses and intervention length warrants further research on optimal dosing strategies.

Magnesium

Magnesium is likely one of the more widely utilized dietary supplements for sleep promotion. With regards to sleep, magnesium has a number of essential functions that include; regulation of gamma-aminobutyric acid (GABA), which can have calming effects on the nervous system, increases in melatonin production, muscle relaxation, regulation of stress responses, regulation of calcium levels which promote muscle contraction and relaxation, and anti-anxiety effects. Furthermore, low magnesium status has been associated with chronic inflammatory stress in both animal and human research studies. Foods rich in magnesium include nuts and seeds, legumes and beans, whole grains, green leafy vegetables, fatty fish like salmon, mackerel, halibut, avocado, bananas, and tofu.

In two randomized control trials that included magnesium supplementation and sleep quality outcomes, both had lackluster results. In one study, 100 healthy adults were randomly assigned to two groups (matched for age, sex, and sleep quality) to receive 320 mg magnesium per day as magnesium citrate or sodium citrate placebo for seven weeks. All study participants reported an increase in sleep quality and there was no difference between groups. Interestingly, serum magnesium did not increase significantly in either group, but when only participants with low serum magnesium were included, magnesium supplementation did significantly increase serum magnesium concentration. The study's results suggest further exploration on the association between magnesium status and sleep quality.[13] In another study, that focused on magnesium supplementation for individuals suffering from nocturnal leg camps on sleep quality, there were no differences in nocturnal leg cramping or quality of sleep. In this study, 94 individuals (mean age 64.9 years) were randomly assigned to receive magnesium oxide or a placebo. All participants experienced a reduction in nocturnal leg cramps, but no improvements in sleep quality were observed.[14]

Despite its popularity, there seem to be few clinical benefits to

magnesium supplementation among individuals with adequate serum magnesium. There is a potential for magnesium supplementation among individuals with low serum magnesium to improve sleep quality.

Zinc

Zinc is a mineral obtained from the diet from foods such as meat, seafood, legumes and beans, nuts and seeds, dairy products, whole grains, vegetables like spinach, mushrooms, asparagus and broccoli, and some fortified breakfast cereals. In relation to sleep, zinc plays a role in the synthesis and regulation of melatonin, which is a hormone critical to sleep-wake cycles. In addition, zinc's influence on neurotransmitter pathways in the brain, especially those related to GABA, which promote relaxation and anxiety reduction can aid in sleep. Indirectly, zinc has anti-inflammatory and immunity properties that also contribute to better sleep through reducing inflammation and reducing illnesses that may lead to sleep disruption.

In the past 10 years, just two randomized control trials have tested the impacts of zinc supplementation alone on sleep outcomes.[15,16] Jafari et al, test zinc supplements (30 mg/day elemental zinc for 12 weeks) against a placebo among college aged women diagnosed with premenstrual syndrome (n = 60).[16] This study found marginal improvements in sleep quality, as measured by the Pittsburgh Sleep Quality survey but the results were not statistically significant. Gholipour et al examined the effect of zinc supplementation on sleep quality among intensive care unit nurses.[15] Nurses were randomized to supplementation (220 mg zinc sulfate) or placebo, and after one month there were statistically significant improvements in sleep quality among the zinc group.

Clinically, more evidence is needed on the impacts of zinc on male subjects, as both studies were predominantly female (100%[16] and 92%[15]). Neither study reported any negative side effects of zinc supplementation at 30 mg/day of elemental zinc[16] or 220 mg of zinc sulfate[15] every 72 hours). Given the existing evidence, there is potential for zinc sulfate to improve sleep quality among healthy females.

Vitamin D

Given the prevalence of vitamin D deficiency in various populations, researchers are increasingly exploring its potential role in sleep regulation. Optimizing vitamin D status through supplementation or increased exposure to sunlight, a natural source of vitamin D may offer a simple and cost-effective strategy to improve sleep quality.

The potential mechanism through which vitamin D impacts sleep involves its regulatory influence on neurotransmitters and hormones implicated in the sleep-wake cycle. Vitamin D receptors are found in the brain's regions involved in sleep regulation, suggesting a direct role. Additionally, vitamin D may modulate serotonin synthesis, a neurotransmitter associated with mood and sleep. One significant aspect of the mechanism is vitamin D's ability to stimulate tryptophan hydroxylase-2 (TPH-2) which plays a pivotal role in the conversion of tryptophan to serotonin.[17] This cascade ultimately leads to the production of melatonin, a hormone crucial for regulating the sleep-wake cycle.

In a study by Majid et al,[18] researchers investigated the impact of vitamin D supplements on sleep in 20-50-year-olds with sleep disorders. The randomized controlled trial involved a supplement group and a control group. The supplement group received vitamin D3 at a dose of 50, 000 IU bi-weekly for eight weeks. The Pittsburgh Sleep Quality Index (PSQI) was utilized to assess sleep quality and disturbances.

Findings revealed a significant improvement in the PSQI score among the supplement group compared to the control group. The supplement group exhibited enhanced sleep quality, reduced sleep latency, and decreased sleep disturbances. These improvements were attributed to the potential role of vitamin D in regulating melatonin, a hormone associated with sleep.

Clinical implications of these findings suggest that vitamin D supplementation may be a valuable adjunctive approach to managing sleep disorders. The dose of 50,000 IU bi-weekly for eight weeks proved effective, though individual variations and underlying health conditions should be considered. Given the prevalence of vitamin D deficiency in various populations, optimizing vitamin D status could be a simple and cost-effective strategy to improve sleep quality, contributing to overall well-being.

L-Theanine

L-theanine, an amino acid primarily found in green tea, has gained attention for its potential impact on sleep due to its calming properties. The proposed mechanism for L-theanine's influence on sleep lies in its ability to modulate neurotransmitters. L-theanine readily crosses the blood-brain barrier and is known to increase the production of gamma-aminobutyric acid (GABA), a neurotransmitter with calming effects. GABA, in turn, helps regulate neuronal excitability, potentially promoting relaxation

and aiding in the initiation of sleep.[19]

In a randomized controlled trial conducted by Hidese et al,[19] the researchers investigated the effects of L-theanine administration on stress-related symptoms and cognitive functions in healthy adults. The Pittsburgh Sleep Quality Index (PSQI) was utilized to assess sleep quality and disturbances. Participants received either a placebo or 200 mg of L-theanine to take daily. The population investigated comprised healthy adults, providing a baseline for understanding the general applicability of L-theanine supplementation.

The results revealed a significant improvement in the PSQI score within the supplement group when contrasted with the control group. The study's outcomes demonstrated that the administration of L-theanine resulted in substantial decreases in stress-related symptoms and notable enhancements in cognitive function when compared to the placebo. Clinically, the implications suggest that L-theanine supplementation may be a suitable nutraceutical ingredient for improving mental conditions and sleep quality in a healthy population. Further research is warranted to explore the direct relationship between L-theanine supplementation and sleep outcomes in diverse populations, but these initial findings provide a promising foundation for future investigations into the role of L-theanine in sleep health.

In summary, the impacts of a number of dietary supplements on sleep quality are supported by potential physiologic mechanisms; however, the actual impact on sleep quality outcomes is highly variable. Continued research in this area is needed to identify appropriate dosages as well as impact on sleep across diverse populations. Physicians and other lifestyle medicine practitioners should consider holistic assessments of dietary intake and other factors that might impact sleep.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article. AJLM

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